

Q: 2(i) (or)
A: 2 (i) (or)

(a) 0.0067
2 significant figures { having 0 doesn't count significant figure }

(b) 30600 J
3 significant figures.

(c) 4.5678

4.57

Q: 2(ii) (or)
A: 2(ii) (or)

- (i) 10^6 ; mega (M)
- (ii) 10^{-3} ; milli (m)
- (iii) 10^{-9} ; nano (n)

(b) 5 km = 5×10^3 m = 5000 m .

Q: 2(iii)
A: 2(iii)

(i) Distance :- 300m + 100m
400m.

(ii) Displacement :- 300m - 100m
200m

(iii) Total speed :- 20s + 10s
30s

$$\text{Average speed} = \frac{v \times s}{t} = \frac{400 \text{ m}}{30 \text{ s}}$$

$$13.3 \text{ m/s}$$

Q.2(iv) _____
A.2(iv) _____

Newton second law:-

The acceleration of an object is directly proportional to the magnitude of the net force in same direction, is inversely proportional to the mass of an object.

Equation (derived) :-

$$a \propto m$$

$$a \propto \frac{1}{\text{mass}}$$

$$= \frac{a}{m}, a = \frac{F}{m}$$

$$F = ma$$

$$Q: 2(v)(or)$$

$$A: 2(v)(or)$$

$$m = 0.15 \text{ kg}$$

$$\text{velocity} = 20 \text{ m/s}$$

$$\Delta p = m(v-u)$$

$$= 0.15(-20 \text{ m/s} - 20 \text{ m/s})$$

$$= 0.15(-40 \text{ m/s})$$

$$= -6 \text{ kg} \cdot \text{m/s} \quad (6 \text{ is magnitude})$$

$$= -6 \text{ kg} \cdot \text{m/s} \quad (6 \text{ magnitude})$$

Change in impulse

$$\text{impulse} = \text{change in momentum}$$

$$= 6 \text{ N} \cdot \text{s}$$

$$Q: 2(v_i)(or)$$

$$A: 2(v_i)(or)$$

Principle of moments:

When a body is in equilibrium, the clockwise moment is equal to the anticlockwise moment about the same pivot.

$$\text{Anticlockwise moment} = \text{clockwise moment}$$

$$4 \text{ N} \times 30 \text{ cm} = 6 \times d$$

$$120 \text{ N} = 6 \times d$$

$$d = \frac{120}{6}$$

$$d = 20 \text{ cm}$$

Q.2(vii)

A.2(vii)

Centripetal force:-

the net force directed towards the centre of a circle that keeps a body moving along a circular path.

Formula:-

$$F_c = \frac{mv^2}{r}$$

① Numerical:-

$$F_c = \frac{mv^2}{r}$$

$$= \frac{0.5 \times (4)^2}{1}$$

$$= \frac{0.5 \times 16}{1}$$

$$= \frac{8}{1} \Rightarrow 8 \text{ N}$$

Q:2(viii)

A:2(viii)

Hooke's Law :-

within the elastic limit, the extension produced in a spring is directly proportional to the applied force ($F = kx$)

Spring constant :-

the force that needed to produce unit extension, its SI unit is N/m .

$$k = \frac{F}{x}$$

$$\frac{12}{0.04}$$

$$= 300 N/m \quad (4cm = 0.04m)$$

Q:2(in)

Q:2(in)

Pressure :- The force acting normally (perpendicularly) per unit area of a surface

$$P = \frac{F}{A} \quad ; \text{ SI unit pascal (Pa) } \\ \text{where, } 1 \text{ Pa} = 1 N/m^2$$

$$P = \frac{F}{A} \Rightarrow \frac{600N}{1.5m^2} = 400 Pa / 400 N/m^2$$

Q: 2(n) (or)
A: 2(n) (or)

Power :-

Rate of doing work (work done per unit time); $P = \frac{W}{t}$; SI unit = watt (W) where $1 \text{ W} = 1 \text{ J/s}$

$$P = \frac{W}{t} = \frac{6000 \text{ J}}{30 \text{ s}} = 200 \text{ W.}$$

Q: 2(ni)
A: 2(ni)

- ① observation of a problem or phenomenon
- ② forming a hypothesis
- ③ Performing experiments to test the hypothesis
- ④ Analyze the results and drawing a conclusion.
- ⑤ If confirmed, forming a theory and law.

Long questions

Q: 3(i)

A: 3(i)

Acceleration:-

Rate of change of velocity of a body.

formula: $a = \frac{v_f - v_i}{t}$

SI unit - m/s^2

(a) $a = \frac{v_f - v_i}{t}$

$$\frac{30 - 10}{8}$$

$$= \frac{20}{8}$$

$$= 2.5 m/s^2$$

(b) distance: $S = \frac{v_i + v_f}{2} \times t$

$$= \frac{10 + 30}{2} \times 8$$

$$= \frac{40}{2} \times 8$$

$$\frac{20}{40} \times 8$$

$$20 \times 8$$

$$160 \text{ m}$$

$$g: 3(ii)$$

$$A: 3(ii)$$

Momentum :-

the quantity of motion of a body, equal to mass $\times$ velocity = ~~mv~~ $p = mv$. It is a vector quantity

~~Law of~~

$$p = m_1 v_1 + m_2 v_2$$

$$1500 \times 12 + 1000 \times 0$$

$$18,000 + 0$$

$$18,000 \text{ Kg.m/s}$$

masses combined :-

$$18000 : 1000 \neq 1500 \times v$$

$$18000 = 2500 \times v$$

$$v = \frac{18000}{2500}$$

$$v = 7.2 \text{ m/s}$$

Q3(ii)(or)

A.3(iii)(or)

Derivation of $P_k = \rho gh$:-

Consider a liquid column of height h and base area A . Volume = $A \times h$
mass = density $\times$ volume.

weight (force on base) = mass $\times g$

$$\rho Ahg$$

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{\rho Ahg}{A} = \rho gh$$

$$P = \rho gh$$

$$1000 \times 10 \times 5 = 50000 \text{ Pa}$$

Total pressure :- Atmospheric + liquid

$$= 1.0 \times 10^5 + 0.5(10)^5$$

$$= 1.5 \times 10^5 \text{ Pa.}$$

Q.3(iv)(or)

A: 3(iv)(or)

mass :- 3 , height :- 5.

(a) PE at top :- mgh

$$= 3 \times 10 \times 5$$
$$= 150 \text{ J}$$

(b) At ground all PE has become ^{me} KE
:- $KE = 150 \text{ J}$

$$150 = \frac{1}{2}mv^2$$

$$150 = \frac{1}{2}(3)v^2$$

$$\frac{150 \times 2}{3} = v^2$$

100

$$\frac{300}{3} = v^2$$

$$\sqrt{100} = \sqrt{v^2}$$

$$10 = v$$

law of conservation of energy -

energy can neither be created nor be destroyed ; it only changes from one form to another, so the total energy of a system stays constant.

MCQs :-

1. B
2. B
3. B
4. A
5. B
6. ~~B~~ ~~B~~ A
7. C
8. C
9. B
10. C
11. B
12. C